

# Conversation with ChatGPT: Recirculating Showers for Vans

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## Prompt:

infinity shower isn't available in israel because they only allow certified installers to handle it.  
how do i make my own recirculating shower?

## Response:

A reliable DIY recirculating shower is basically **four subsystems**:

### 1. Shower loop

Water path:

```
None
showerhead
↓
shower pan drain
↓
hair / debris filter
↓
circulation pump
↓
sediment filter
↓
carbon filter
```

↓  
UV sterilizer  
↓  
return line  
↓  
showerhead

The system runs in a **closed loop while showering**, then **switches to normal drain** when finished.

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## Core components

### Pump

You want a **quiet 12V diaphragm pump**.

Good spec:

- **4–6 L/min flow**
- **30–60 PSI**

Common choices:

- Shurflo 4008
  - Seaflo 42 series
- 

### Filter stack (3 stages minimum)

#### 1 — hair filter

- stainless shower drain strainer

#### 2 — sediment

- 5 micron cartridge

#### 3 — carbon

- removes soap + odor

Standard **10" water filter housings** work well.

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## UV sterilizer

Kills bacteria in reused water.

Look for:

- **12V UV-C**
  - **254 nm**
  - rated **2–4 L/min**
- 

## Diverter valve

You need **3-way plumbing**:

Mode 1:

None

normal shower → grey tank

Mode 2:

None

recirculation loop

Manual **3-way ball valve** works fine.

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## Shower pan

Important:

The **drain must collect water fast** or the pan floods.

Use:

- **90 mm shower drain**
  - straight drop to filter box
- 

## Basic plumbing layout

```
None
showerhead
  ↑
  |
UV sterilizer
  ↑
carbon filter
  ↑
sediment filter
  ↑
pump
  ↑
hair trap
  ↑
drain pan
```

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## Water usage

Typical recirculating system:

- **5–8 liters total water**
  - unlimited shower time
- 

## Maintenance (very important)

Every shower:

- rinse hair trap

Every ~10 showers:

- clean sediment filter

Every 2–3 months:

- replace carbon filter

UV bulb:

- yearly
- 

## Things that make these systems fail

From vanlife builds:

### 1. weak filtration

→ smells fast

### 2. pump too strong

→ pan floods

### 3. oils / conditioners

→ filters clog

### 4. heat loss

→ shower gets cold

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## Key trick most DIY systems miss

Add a **small recirculation reservoir (5–10L tank)** between the drain and pump.

Benefits:

- prevents pump cavitation
  - stabilizes flow
  - captures hair
- 

## Power draw

Typical:

- pump: **50–80W**
- UV: **10–15W**

Total:

**~60–90W**

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## If I were building your van system

I would use:

- **8L recirculation tank**
- **Shurflo pump**
- **10" triple filter housing**
- **12V UV sterilizer**
- **1.5 GPM shower head**

This is basically **what commercial systems do internally**.

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If you want, I can also show you the **very important design improvement almost no DIY recirculating showers use** that makes them **10× cleaner and far less maintenance**, and is probably what you want for **commercial vans**.

## Prompt:

the hair and debris filter needs to be able to clean itself off into the greywater jerrycans without anyone having to touch it, take it out, or look at it or anything like that

## Response:

You want a **self-purging solids separator** before the recirculation loop. The clean way to do it in vans is a **two-stage drain separator with an automatic purge line**.

## Architecture

```
None
shower drain
↓
vortex / cyclone separator
↓
fine screen filter
↓
recirculation tank
↓
pump → filters → UV → showerhead

separator purge port
↓
greywater jerrycans
```

## Stage 1 — vortex separator (self-cleaning)

Use a **cyclone / swirl chamber**:

None

tangential inlet

→ spinning water

→ hair + debris forced outward

→ fall to bottom sludge chamber

Bottom chamber connects to a **continuous purge line** that drains into the grey jerrycans.

Because the flow is constantly flushing the bottom, **solids never accumulate in the loop**.

Common parts to adapt:

- **hydrocyclone sediment separator**
- **spa / pool vortex prefilter**
- **centrifugal inline solids separator**

These handle hair extremely well.

## Stage 2 — wedge-wire screen (self-flushing)

After the cyclone:

Install a **coarse wedge-wire screen (~300–500 micron)**.

Key detail:

None

main flow → recirculation loop

small bleed flow → greywater line

That small bleed constantly washes debris off the screen.

This is exactly how **industrial self-cleaning filters** work.

## Purge plumbing

The purge line should be **gravity drain** to the grey jerrycans.

Add:

- check valve
- air gap or vent

So greywater cannot backflow.

## Why this works

Hair is difficult because it **wraps around filters**.

The cyclone stage solves this because hair:

- is light
- tangles
- migrates to outer vortex wall
- drops into purge chamber

Industrial systems separate **fibers and hair** this way.

## Tank before the pump

After filtration:

Use a **small buffer tank (5–8L)**.

Purpose:

- stabilizes flow
- keeps pump from cavitating
- allows purge water to leave system

None

drain

↓

cyclone

↓

screen

↓

buffer tank

↓

pump

# Greywater purge rate

Typical design:

- **5–10% of flow purged**

Example:

None

shower flow 5 L/min

purge 0.25–0.5 L/min

That continuously carries hair away.

## Result

User experience:

- shower normally
- debris automatically flushed
- nothing to clean or touch
- greywater jerrycan collects solids

Exactly what you want for **commercial vans**.

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If you want, I can also show you the **three sanitation problems that kill most recirculating showers after a few months**, and the engineering fixes that make them **actually viable for long-term heavy use**.

**Prompt:**

yes pls

## Response:

Three failure modes show up again and again in long-term recirculating shower use.

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# 1. Biofilm (the biggest problem)

Warm water + soap = **bacteria slime** in hoses and filters.

Once biofilm forms:

- smell appears
- filters clog faster
- UV becomes less effective

## Fix

Design the system so it **fully drains and dries** after every shower.

Key features:

None

loop drain valve → greywater  
sloped plumbing  
no standing water

Also add:

- **UV-C sterilizer**
- optional **ozone injector** or periodic **H<sub>2</sub>O<sub>2</sub> flush**

Many commercial systems fail because water sits in the loop.

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# 2. Soap buildup

Conditioners, oils, and thick soaps **destroy filtration**.

Symptoms:

- filters clog quickly
- water gets cloudy
- smell develops

**Fix**

Use **low-residue soap only**.

Best options:

- Dr Bronner type soap
- camping biodegradable soaps
- no conditioners

Engineering fix:

Add a **carbon filter after sediment filter**.

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## 3. Temperature loss

Water cools quickly because it constantly circulates through pipes and filters.

Typical drop:

None

1–2°C per minute

After 10 minutes the shower can feel cold.

**Fix**

Add **small inline heater or heat exchanger**.

Two common solutions:

**A — hot water heater mixing**

None

recirc loop

↓

thermostatic mixing valve

↓

hot water heater

Only a little hot water enters to maintain temp.

### **B — inline electric heater**

Example spec:

- 12V or 24V
- 300–600W

This compensates for heat loss.

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## **Other design details that matter**

### **Use large plumbing**

Minimum:

None

1/2" lines

Small tubing restricts flow and causes pump strain.

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### **Pump placement**

Pump should sit **after the buffer tank**, not directly on the drain.

None

drain

↓

separator

↓

buffer tank

↓

pump

Prevents cavitation.

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### Filtration stack (ideal)

None

hair separator

↓

200–500 micron screen

↓

20 micron sediment

↓

5 micron sediment

↓

carbon

↓

UV sterilizer

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## Water volume sweet spot

Use **6–10 liters** in the loop.

Too small:

- temperature swings
- pump cavitation

Too large:

- sanitation harder

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## Power use

Typical DIY system:

None

pump 50–80W

UV 10–15W

heater 300–600W (optional)

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## One more trick that dramatically improves hygiene

Add a **30-second fresh water rinse cycle** at the end of the shower.

Flow:

None

fresh water tank

↓

loop flush

↓

greywater

This removes soap and body oils from the loop.

It makes filters last **3–5× longer**.

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If you're building vans for **heavy personal + commercial use**, I can also show you the **professional architecture used in expedition vehicles and yachts** that makes recirculating showers extremely reliable and much easier to maintain.

## Prompt:

i wanna use a pump with way higher flow tho, i want a high flow shower with a rainfall showerhead...

## Response:

You can, but the system design must scale with it. High-flow recirculating showers are harder mainly because **filtration, separation, and heating all scale with flow**.

## Target specs for a “luxury” rainfall system

Typical rainfall heads:

- **8–12 L/min (2–3 GPM)**
- some even **15 L/min**

Design around **10–12 L/min** to be safe.

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## Key changes vs normal DIY systems

### 1. Much larger buffer tank

Small loops fail at high flow.

Use:

**15–25 L recirculation tank**

Purpose:

- prevents pump starvation
- stabilizes temperature
- allows debris to settle

None

drain

↓

cyclone separator

↓

screen filter

↓

buffer tank (20L)

↓

pump

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## 2. High-flow pump

Instead of RV pumps:

Look for **marine pressure pumps**.

Good spec:

- **12–15 L/min**
- **40–60 PSI**
- 12V or 24V

Common examples:

- **Shurflo 5.7**
- **Seaflor 55 series**
- **Jabsco PAR-Max**

These can run rainfall heads easily.

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### 3. Bigger plumbing

Small tubing kills rainfall showers.

Use:

None

3/4" main lines

1/2" minimum to shower

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### 4. Larger filters

Standard 10" filters restrict high flow.

Use:

**20" "Big Blue" filter housings**

Stack:

None

sediment 50 micron

sediment 5 micron

carbon

These support **15–20 L/min** without restriction.

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### 5. Stronger UV sterilizer

UV must match flow.

For rainfall system:

- **10–15 L/min UV sterilizer**

Small 2–4 L/min units won't work.

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## 6. Heating becomes critical

High flow means **heat loss increases dramatically**.

Best solution in vans:

**diesel hydronic heater heat exchanger**

None

recirculating loop

↓

plate heat exchanger

↓

diesel heater loop

This keeps water **perfectly hot indefinitely**.

Electric heaters struggle at these flow rates.

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## 7. Shower pan drain size

You must avoid pan flooding.

Use:

**90 mm RV shower drain**

Minimum.

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## Expected performance

With a well-designed system:

- 10–12 L/min rainfall shower
- 15–20 L water loop
- virtually unlimited shower time

Feels like a **normal luxury home shower**.

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## One more thing most people miss

Rainfall showers require **high pressure stability**.

To achieve that:

Install a **small accumulator tank** after the pump.

```
None
pump
↓
accumulator tank
↓
filters
↓
UV
↓
shower
```

This removes pulsing and makes the rainfall feel smooth.

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If you want, I can also show you a **recirculating shower architecture used in expedition trucks and research vessels** that supports **hotel-level rainfall showers while using only ~20 liters of water total**, and it's much more robust than typical vanlife builds.

## Prompt:

what kind of water tester should i buy to test the hygienic cleanliness of the filtered water as it's being turned into greywater and then reused?

## Response:

Don't buy just **one** tester.

For a recirculating shower, the best practical stack is:

### 1. **ATP water tester** for fast day-to-day hygiene checks

This is the closest thing to a quick "is this getting biologically gross?" field check. Hygiena's **AquaSnap / EnSURE Touch** style system is made for rapid ATP testing in water, with results in seconds.

### 2. **Turbidity meter** for clarity / filter performance

If turbidity rises, your filtration is slipping even before the water obviously looks bad. Portable meters like Hach's **2100Q** are purpose-built for field water testing.

### 3. **Periodic bacteria test kit** for actual coliform / E. coli screening

ATP is useful, but it does **not** replace microbiology testing. For periodic verification, use something like **IDEXX Colilert / Colilert-18**, which is specifically for detecting total coliforms and **E. coli**.

So the real answer is:

- **Best single thing to buy first: an ATP water tester**
- **Best serious setup: ATP tester + turbidity meter + periodic Colilert tests**

Also important: **don't waste money on a TDS meter** for this purpose. TDS and pH can be useful for general water characterization, but they do **not** tell you whether reused shower water is hygienically clean.

For your use case, I'd do this:

- before each day / between users: **ATP**

- whenever tuning filters: **turbidity**
- on a schedule and after design changes: **coliform/E. coli test**

If you want the simplest buying answer: **start with a Hygiena water ATP kit**. That's the most relevant first purchase.